

Matrix Spencer, twice: a comparison of arXiv:2608.28816 and arXiv:2609.15025

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Abstract

This note compares two recent arXiv preprints on the Matrix Spencer conjecture: *A Proof of the Matrix Spencer Conjecture* by Emrullah Akbas and the author (arXiv:2608.28816, posted 28 August 2026), and *Matrix Spencer: Eight Standard Deviations Suffice and an Almost-Linear Time Algorithm for Dense Input* by Zhao Song and Lichen Zhang (arXiv:2609.15025, posted 14 September 2026).

This report is by no means intended to be exhaustive: the examples discussed below represent only a subset of the technical and conceptual overlap with Akbas and Sra that appears in Song and Zhang *without* attribution. In particular, we focus on the most substantive points of overlap, rather than attempting to identify every passage, construction, or technical ingredient in Song and Zhang that closely parallels material already present in Akbas and Sra. The comparison shows that the same barrier family, the same diagonal-to-full interpolation of a log-partition function idea, and the same matrix-weighted Poincaré pipeline is used in Song and Zhang *without* attributing to Akbas and Sra. It also identifies what is genuinely new in the second paper. The note closes with the corrections the record requires. Every quotation and every section, definition, and lemma number below is taken from the posted v1 of each paper and can be checked against it.

0 TL;DR

Our report is by no means exhaustive; it highlights only some of the main instances in which material from our work appears in Song and Zhang without proper attribution:

- (1) The authors of [SZ26] claim that the Matrix Spencer “[...] conjecture was known only under rank, block-diagonal, or Frobenius-norm restrictions.” and “We prove it: a signing of discrepancy below $8\sqrt{n}$ always exists.”, see abstract of [SZ26].
- (2) The authors of [SZ26] write in their abstract that “The existence proof is a partial-coloring argument with one new estimate: a hereditary Gaussian small-ball bound for the spectral body $\{x \in \mathbb{R}^n : \|\sum_i x_i A_i\| \leq R\}$, proved by interpolating a log-partition function from a diagonal model to the noncommutative one under a matrix-weighted Poincaré inequality.”
- (3) The authors of [SZ26] write at page 16 “Independent and Concurrent Work. After completing a draft of this paper, the authors became aware of the independent and concurrent work of [AS26], which proves a similar result. The proof strategy of this paper is fundamentally different from [AS26].”

Against these statements, we make the following observations:

- (1) At the time [SZ26] was posted, the Matrix Spencer conjecture had already been resolved by [AS26], whose v1 appeared on 28 August 2026. Thus, the statements that the conjecture “was known only under rank, block-diagonal, or Frobenius-norm restrictions” and that “the general case has remained open” do not accurately describe the state of the literature at the time of posting. [AS26] states and proves the full Matrix Spencer conjecture as Theorem 1.2.
- (2) The hereditary Gaussian small-ball estimate presented in [SZ26] as “one new estimate” has a direct antecedent in [AS26]: Theorem 1.3 of [AS26] is explicitly titled *Hereditary Matrix Small Ball*. More importantly, the existence proof in [SZ26], based on “[...] interpolating a log-partition function from a diagonal model to the noncommutative one under a matrix-weighted Poincaré inequality,” follows essentially the same proof architecture as [AS26]. Indeed, [SZ26] uses in Section 4 the same parameterized barrier family, the same diagonal-to-full interpolation of a log-partition function, and the same matrix-weighted Poincaré/resolvent pipeline. These are not merely high-level similarities: the principal constructions and analytic mechanism coincide under a change of notation, in particular, the parameterized barrier family and diagonal-to-full interpolation agree exactly after relabeling. The additional machinery used by [SZ26, Appendix A] to sharpen the discrepancy constant below 8 refines the original argument, but it remains built on the same underlying barrier/interpolation/small-ball/Poincaré architecture of [AS26]. In particular, Appendix A of [SZ26] replaces or strengthens individual components of framework by [AS26] rather than introducing a novel route to Matrix Spencer.
- (3) In light of point (2) above, the claim that the two proof strategies are “fundamentally different” is not supported. See also the technical comparison below. In particular, the two arguments use the same barrier family, the same diagonal-to-full interpolation, and the same matrix-weighted Poincaré mechanism, up to notation and quantitative refinements.

Below, we highlight several more specific instances of overlap that, to our knowledge, are not attributed by Song and Zhang [SZ26] to Akbas and Sra [AS26], although the corresponding material already appears in [AS26]. We have used AI tools to assist with the comparison and assessment below. Accordingly, our exposition may contain errors or omissions; in case of any discrepancy, the original sources should be treated as authoritative, and any claim concerning [AS26] should be checked directly against [AS26]. Responsibility for this report remains with its authors.

1 Chronology and claims

The public record consists of two arXiv postings.

28 Aug 2026	[AS26] v1. Theorem 1.2 (<i>Matrix Spencer</i>): for symmetric $n \times n$ matrices $A_1, \dots, A_n$ with $\ A_i\ _{\text{op}} \leq 1$, $\min_{x \in \{\pm 1\}^n} \ \sum_i x_i A_i\ _{\text{op}} \leq C\sqrt{n}$; “moreover, a coloring with discrepancy at most $C\sqrt{n}$ can be found efficiently.” Theorem 1.3 (<i>Hereditary Matrix Small Ball</i>) is the engine; the abstract singles it out as “a hereditary small-ball estimate for Gaussian series that should be of independent interest.”
14 Sep 2026	[SZ26] v1 (title page dated 1 September 2026). Existence of a signing with discrepancy below $8\sqrt{n}$; a randomized $n^{3+o(1)}$ polylog($1/p$)-time algorithm with discrepancy below $12\sqrt{n}$.

What [SZ26] says about the state of the problem. The abstract: “the conjecture was known only under rank, block-diagonal, or Frobenius-norm restrictions. We prove it.” The introduction, after listing the restricted results of Hopkins–Raghavendra–Shetty, Dadush–Jiang–Reis, and Bansal–Jiang–Meka: “The general case has remained open.” The abstract again: “The existence proof is a partial-coloring argument with one new estimate: a hereditary Gaussian small-ball bound.” Section 2.1 repeats it: proving the conjecture “needs one new estimate inside a known frame.”

What [SZ26] says about [AS26]. The paper cites [AS26] exactly twice. Once in the bibliography, and once in the section “Acknowledgments & AI Disclosure” on page 16:

“Independent and Concurrent Work. After completing a draft of this paper, the authors became aware of the independent and concurrent work of [AS26], which proves a similar result. The proof strategy of this paper is fundamentally different from [AS26].”

[AS26] does not appear in the abstract, in the introduction, or in Section 1.2 (“Previous work”), which discusses the restricted results at length. The same page discloses that the authors “have started to prompt AI to get a proof for matrix Spencer” on 28 July 2026, “obtained the existence proof on August 20, 2026,” and that “the proof of the existence of such a signing comes mainly from AI.”

Two statements that are false as posted. On 14 September 2026 the general case had not “remained open”; it had been known publicly for seventeen days (and privately to us, even longer), and the conjecture was not “known only under” restrictions. [SZ26] cites [AS26], so both sentences were posted with knowledge of the result that contradicts them. The rest of this note concerns the third statement, that the proof strategy is “fundamentally different,” and the fourth, that the hereditary small-ball estimate is “new.”

2 Side-by-side comparison of the proofs

Here we highlight how even on the level of details, the work by [SZ26] uses ideas of [AS26] to prove the Gaussian measure lower bound as required by Gluskin’s partial coloring framework, and, unfortunately without attributing so. Here we compare [AS26] Section 3 against [SZ26] Section 4, and it goes as follows.

2.1 The barrier. [AS26] Section 3.1 defines, for an integer q and $\alpha \geq \beta > 0$,

$$v_{q,\alpha,\beta}(t) = \beta \ell(t) + (\alpha - \beta) r_q(t), \quad \ell(t) = -\log(1 - t^2), \quad r_q(t) = \sum_{k \geq q} \frac{t^{2k}}{k}, \quad (1)$$

and the spectral potential $V(B) = \text{Tr } v_{q,\alpha,\beta}(B)$ for $\|B\|_{\text{op}} < 1$, $V(B) = +\infty$ otherwise (their (3.4)). [SZ26] Definition 4.2 (“Barrier potential”) defines, for an integer $\tau \geq 2$ and $\kappa_1 \geq \kappa_0 > 0$,

$$\phi(s) = \kappa_0 \phi_0(s) + (\kappa_1 - \kappa_0) \phi_{1,\tau}(s), \quad \phi_0(s) = -\log(1 - s^2), \quad \phi_{1,\tau}(s) = \sum_{k \geq \tau} \frac{s^{2k}}{k}, \quad (2)$$

extended by $+\infty$ outside $(-1, 1)$, with $\Phi(X) = \text{tr } \phi(X)$. Under

$$q = \tau, \quad \alpha = \kappa_1, \quad \beta = \kappa_0,$$

(1) and (2) are the same function, the constraint $\alpha \geq \beta$ is the constraint $\kappa_1 \geq \kappa_0$, and V is Φ . This is an identity of parameterized functions, not a resemblance between two logarithmic barriers.

The design rationale is also the same: weight the low-degree part of $-\log(1-t^2)$ lightly and the high-degree tail heavily, so that curvature concentrates at the spectral boundary while the bulk is barely penalized ([AS26] Section 3.1, second paragraph; [SZ26] Section 2.1, “The hereditary small-ball estimate”).

2.2 The diagonal-to-full interpolation. [AS26] equations (3.2)–(3.3): fix an orthonormal basis, split $B(g) = B_d(g) + B_o(g)$ into diagonal and off-diagonal parts, and set

$$B_t(g) = B_d(g) + tB_o(g), \quad Z_t = \mathbb{E}_{\gamma_n}[e^{-V(B_t(g))}], \quad F(t) = \log Z_t, \quad 0 \leq t \leq 1.$$

[SZ26] Definition 4.4 (“Diagonal-to-full interpolation”): fix an orthonormal basis, write $A = A_{\text{diag}} + A_{\text{off}}$, and set

$$A_t = A_{\text{diag}} + tA_{\text{off}}, \quad a_t = \mathbb{E}[e^{-\Phi(A_t)}], \quad a(t) = \log a_t, \quad t \in [0, 1].$$

Same object, same role: bound the diagonal endpoint directly, then control the second derivative of the log-partition function along the path.

2.3 The analytic pipeline. The remaining steps correspond one to one.

Mechanism	[AS26]	[SZ26]
Matrix-weighted Poincaré curvature criterion	Section 2.1, Lemma 2.3 (<i>Matrix-weighted curvature criterion</i>); Lemma 2.2 gives the Gaussian specialization	Section 4.4, Lemma 4.10 (<i>Matrix-weighted Poincaré criterion</i>)
Spectral boundary/bulk decomposition and Hessian estimates	Section 3.2, in particular Lemmas 3.2–3.6	Section 4.3, in particular Definitions 4.7–4.8 and Lemma 4.9
Curvature lower bound for the one- and two-resolvent matrix weights	Section 3.2, Proposition 3.7 (<i>Curvature lower bound</i>)	Section 4.5, curvature calculation in the proof of Lemma 4.12
Resolvent-weighted Poincaré estimates	Section 3.2, Proposition 3.8	Sections 4.5–4.6, Lemmas 4.12–4.13
Iteration over matrix-valued polynomials / resolvent moments	Section 3.2, Lemmas 3.10–3.11	Section 4.7, Lemma 4.14
Control of the interpolation Hessian	Section 3.3, Lemma 3.13	Section 4.7, Lemma 4.14
Hereditary Gaussian small-ball estimate	Theorem 1.3 (<i>Hereditary Matrix Small Ball</i>)	Lemma 4.1 (<i>Six-regime hereditary small ball</i>)

The vocabulary coincides as well: “hereditary,” “matrix-weighted Poincaré,” “log-partition function,” “interpolation from a diagonal model.”

2.4 What is new in [SZ26]. Three things, and they deserve to be stated plainly.

- (i) *Constants.* [SZ26] carries explicit, certified coefficients through the whole argument (Section 4.8, Appendix A) and reaches discrepancy below $7.8795\sqrt{n}$. [AS26] proves $O(\sqrt{n})$ with an unspecified universal constant.
- (ii) *The hereditary radius.* [AS26] Theorem 1.3 has radius $\kappa\sqrt{n}(1 + \log \frac{2d}{n})^2$; [SZ26] Lemma 4.1 has radius $c_{\text{rad}}\sqrt{m}\ell_{\text{asp}}$ with $\ell_{\text{asp}} = \max\{1, \log(N/m)\}$. Linear rather than squared dependence on the log-aspect-ratio is a genuine sharpening of the estimate.

- (iii) *Running time.* The $n^{3+o(1)}$ algorithm and the sequence of algorithms leading to it (Sections 6–10) have no counterpart in [AS26], which gives a polynomial-time construction.

Improving a constant and a running time are contributions. They are contributions to a theorem that had been proved.

3 Assessment

On “fundamentally different.” The overall/general proof architecture and idea coincides with the one of [AS26]. In particular, Sections 2.1–2.3 show that the same barrier family, the same interpolation, and the same Poincaré pipeline of [AS26] are used by [SZ26], only under a slight change of notation and minor technical details, without attributing them properly to [AS26].

On “one new estimate.” The estimate advertised as new in the abstract of [SZ26] is, up to the quantitative improvements in Section 2.4, Theorem 1.3 of [AS26], titled “Hereditary Matrix Small Ball” and highlighted in the abstract of [AS26] seventeen days earlier.

4 What the record should say

A corrected v2 of [SZ26] should contain at least the following.

- (a) Abstract and introduction: a statement that the conjecture was resolved in [AS26] (arXiv: 2608.28816, 28 August 2026), in place of “remained open” and “known only under ... restrictions,” with the contribution described as improving the constant and the running time.
- (b) Abstract: attribution of the hereditary Gaussian small-ball estimate to [AS26], with “one new estimate” removed. Rather qualifying that the contributions of [SZ26] are in obtaining explicit constants and improved algorithmic guarantees.
- (c) Section 1.2: a substantive comparison of Section 4 of [SZ26] with Section 3 of [AS26], listing the shared barrier, interpolation, and Poincaré machinery and stating specifically what differs, and attributing the general proof idea to [AS26].
- (d) Acknowledgments: deletion of “fundamentally different,” or its replacement by a specific technical claim.

References

- [AS26] E. Akbas and S. Sra. *A Proof of the Matrix Spencer Conjecture*. arXiv:2608.28816, v1 posted 28 August 2026. <https://arxiv.org/abs/2608.28816>
- [SZ26] Z. Song and L. Zhang. *Matrix Spencer: Eight Standard Deviations Suffice and an Almost-Linear Time Algorithm for Dense Input*. arXiv:2609.15025, v1 posted 14 September 2026. <https://arxiv.org/abs/2609.15025>